



- 12 The diagram shows a sector of a circle with centre O and radius 9 cm.



NOT TO SCALE

The perimeter of the sector is $(18 + 2\pi)$ cm.

Find the area of the sector.
Give your answer in terms of π .

$$\frac{\theta}{360} \times 2\pi r + 18 = 18 + 2\pi \quad \theta = 60^\circ$$

$$\frac{\theta}{360} \times 18\pi = 2\pi$$

$$18\pi \theta = 2\pi \times 360$$

$$\frac{18\pi \theta}{18\pi} = \frac{2\pi \times 360}{18\pi}$$

$$\frac{60^\circ}{360^\circ} \times \pi \times 9^2$$

$$\frac{\pi \times 81 \times 60}{360}$$

$$9\pi \text{ cm}^2 \quad \underline{\quad 9\pi \quad} \text{ cm}^2 \quad [4]$$

- 13 These are Rahul's 10 test scores.

9 8 9 10 7 x 9 9 x 7

The mean of these scores is 8.

Find the interquartile range.
You must show all your working.

$$\frac{9 + 8 + 9 + 10 + 7 + 2x + 9 + 9 + x + 7}{10} = 8$$

$$68 + 2x = 80$$

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$$2x = 80 - 68$$

$$\frac{2x}{2} = \frac{12}{2}$$

$$x = 6$$

6, 6, 7, 7, 7, 8, 9, 9, 9, 9, 10

$$L_2 = \frac{10 + 7}{2}$$

$$= 7.5^{\text{th}} \text{ place}$$

$$= \frac{6 + 7}{2} = 6.5$$

$$U_4 = \frac{80 + 9}{2}$$

$$= 8.2^{\text{nd}} \text{ place}$$

$$= \frac{88}{2}$$

$$= 44$$

$$IQR = 9 + 6.5 = 15.5$$

$$\underline{\quad 15.5 \quad} \quad [4]$$